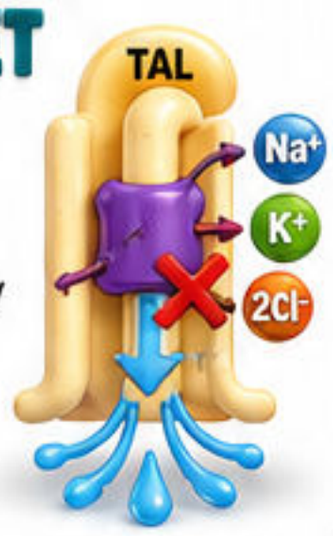


ETHACRYNIC ACID [EDECRIN]

Pharma Talk

Loop diuretic; phenoxyacetic acid derivative.
Potent blocker of NKCC2 in the thick ascending limb.

1 MECHANISM OF ACTION (MOA) & TARGET



- **TARGET:** Na-K-2Cl cotransporter (NKCC2) in the thick ascending limb (TAL).
- **MOA:** Blocks ion reabsorption → ↓ medullary osmotic gradient → profound diuresis.
- **Analogy:** A dam breaking in the kidney's primary concentrating segment.

2 FORMS & STRENGTHS



- **PO tablets:** 25 mg, 50 mg
- **IV vial:** 50 mg (sodium ethacrynate)



3 INDICATIONS (FDA-APPROVED & MAJOR OFF-LABEL)



- **FDA-approved:** Edema secondary to CHF, hepatic cirrhosis, or renal disease, including nephrotic syndrome.
- **Off-label:** Acute pulmonary edema, hypercalcemia management.



4 DOSAGE & ADMINISTRATION

A) ADULT DOSE

- Standard: 50 mg PO daily
- Max: 200 mg/day
- Route: PO or IV
- Loading: N/A
- Maintenance: 50–200 mg/day

B) PEDIATRIC DOSE

- PEDIATRIC USE — Safety/efficacy not established.
- Use with caution; data limited.

C) ADMINISTRATION



- PO: With or without food
- IV infusion slowly over 5 minutes
- Do NOT give as rapid IV push

D) ADJUSTMENTS



- Renal: ↓ dose in severe impairment
- Hepatic: Use cautiously to avoid encephalopathy

5 PHARMACOKINETICS & DYNAMICS (PK/PD)







- **Onset:** PO 30 minutes; IV 5 minutes
- **Peak effect:** PO 1–2 hours; IV 30–60 min
- **Duration:** 4–6 hours
- **T1/2:** 30–60 minutes
- **Metabolism:** Extensive hepatic conjugation
- **Excretion:** Renal (as conjugates)

6 ADVERSE EFFECTS & TOXICITY



- **Common:** Hypokalemia, hypomagnesemia, hyperuricemia, hypotension, metabolic alkalosis
- **Severe:** Permanent ototoxicity (deafness, tinnitus, vertigo)
- Severe dehydration; volume depletion

7 CONTRAINDICATIONS & INTERACTIONS

CONTRAINDICATIONS

- ABSOLUTE CI: ANURIA
- ABSOLUTE CI: SEVERE WATERY DIARRHEA

BOXED WARNING
PROFOUND WATER AND ELECTROLYTE DEPLETION

MAJOR INTERACTIONS



- Aminoglycosides → additive ototoxicity → irreversible deafness



- Lithium → ↓ clearance → increased levels → toxicity

8 SPECIAL POPULATIONS (PREGNANCY & LACTATION)



- Pregnancy: Use only if benefit justifies risk



- Lactation: Avoid breastfeeding entirely



- Geriatric: ↑ risk of hypotension & falls

9 CLINICAL PEARLS



- Only non-sulfonamide loop diuretic—ideal for severe sulfa allergy.
- Highest risk of ototoxicity among loop diuretics—especially with rapid IV push or renal failure.
- Monitor K^+, Mg^{2+}, volume status, and hearing.
- Avoid concurrent ototoxic medications.

10 GLOSSARY OF ABBREVIATIONS



CHF = Congestive Heart Failure



FDA = Food and Drug Administration



NKCC2 = Sodium-Potassium-Chloride Cotransporter



TAL = Thick Ascending Limb